INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

# ME 362 VIBRATIONS I





Lecture 1

Introduction to Vibration, its Modelling, and Related Concepts

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Mechanical Engineering Department, KNUST

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# REFERENCE BOOKS


- Mechanics of Machines, Advanced Theory and Examples, John Hannah, R. C. Stephens.
- Mechanics of Machines, Elementary Theory and Examples, John Hannah, R. C. Stephens.
- Theory of Machines, R.S. Khurmi and J.K. Gupta, 2004
- Vibration with Control, Daniel J. Inman, Wiley, 2006.
- Vibrations in Rotating Machinery, Professional Engineering Publishers, Wiley, 2000
- Vibrations in Rotating Machinery, IMechE, Wiley, 2004.
- VECTOR MECHANICS FOR ENGINEERS (Statics and Dynamics), Ferdinand P. Beer, E. Russell Johnston, Jr., David F. Mazurek, Phillip J. Cornwell, Elliot R. Eisenberg

# Grading:

- Continuous Assessment 30%
- Final Exam 70%

**Prerequisites:** A previous course in:

ME 161-Basic Engineering Mechanics

ME 262/264/260-Theory of Machines

ME 361-Dynamics of Machinery

**Homework:** About 4-7 problems will be assigned after every meeting and these will be collected for grading. Collaboration is encouraged, but independent work is required and essential in order to perform well on the exams.

# COURSE OUTLINE

- Unit 1: Introduction to Vibration, its Modelling, and Related Concepts
- Unit 2: Response to Excitations
- Unit 3: Multiple-Degree-of-Freedom Systems
- Unit 4: Torsional Vibration Systems
- Unit 5: Design for Vibration Suppression
- Unit 6: Vibration Testing and Measurement

# UNIT 1
# Introduction to Vibration, its Modelling, and Related Concepts

# Mechanical vibration

- In ME 361 we studied kinematics and kinetics of rigid bodies separately
- For flexible(non rigid) members the kinematic and kinetic analysis cannot be decoupled
- Flexible members that have mass oscillates when subjected to a force

# Mechanical vibration cont'd...

- Mechanical vibration is repetitive motion of a particle or system of bodies relative to a stationary reference frame or equilibrium position.
- Vibrations occur everywhere in our lives

➢ heart rate

➢ music(sound)

➢ earthquakes

➢Machines and structures etc.

- The performances of many engineering systems are limited by the vibrational properties of the systems. Vibration can be harmful and in such cases it must be avoided. It can be very useful and desired, or may increase the performance of an engineering system.

# Mechanical vibration cont'd...

Present-day machines and structures often contain high-energy sources which create intense vibration excitation problems, and modern construction methods result in systems with low mass and low inherent damping. Therefore careful design and analysis is necessary to avoid resonance or an undesirable dynamic performance.

# Uses of vibration analysis

- Breathing
- Hearing
- Seeing
- Speaking
- walking

- Preventive maintenance
- Incoming inspection
- Quality control
- Engineering and research
- Plant engineering
- Field service and machinery start-up
- Sales and customer satisfaction
- Noise control
- Auscultation

# Some definitions

- **Vibration** is a motion repeated at some intervals. It is expressed in terms of frequency.
- **Cycle** is a single complete repetition of a vibratory motion.
- **Period** is the time required to complete one cycle
- **Frequency** is the number of cycles completed in a given interval of time, usually one second, but occasionally, one minute.
- **Amplitude** is the maximum displacement of a vibrating object from its initial position.
- **Torque** is an action tending to produce rotation of an object.
- **Torsional Vibration** is the twisting and untwisting of a shaft resulting from the periodic application of torque.
- **Damp** is to dissipate energy from a vibrating system.

# Classification of Vibration

- Free and Forced Vibration
- Undamped and Damped Vibration
- Linear and Nonlinear Vibration
- Deterministic and Random Vibration
- Longitudinal and transverse vibrations

# Vibrations analysis

Vibrations analysis can be carried out most conveniently by adopting the following three-stage approach:

I. MODEL

II. EQUATIONS

III. EXCITATION(RESPONSE)

# VIBRATIONS SYSTEM COMPONENTS

A vibration system comprises inertia components, stiffness components, and damping components.



- The inertia components have kinetic energy when the system is in motion. The kinetic energy of a rigid body undergoing planar motion is

$$F = m\ddot{x}; \qquad K.E = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

- A linear stiffness component (a linear spring) has a force displacement relation of the form



$$F = kx; \qquad P.E = \frac{1}{2}kx^2$$

- where *F* is applied force and x is the component's change in length from its unstreched length.
- The stiffness *k* has dimensions of force per length.

# MECHANICAL SYSTEM COMPONENTS

- A *dashpot(damper)* is a mechanical device that adds viscous damping to a mechanical system. A linear viscous damping component has a force-velocity relation of the form;

$$F = cv; \qquad D.E = -\int c\dot{x}\, dx$$

- where $c$, is the damping coefficient of dimensions mass per time.





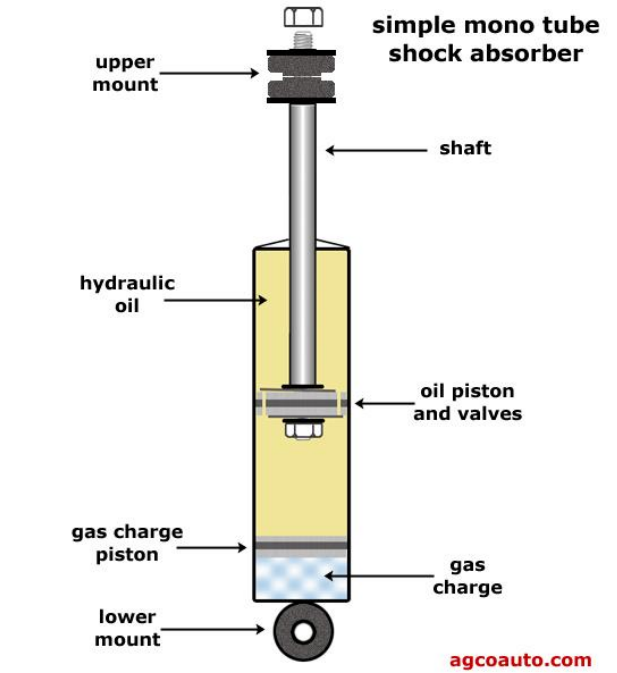


# MECHANICAL SYSTEM COMPONENTS

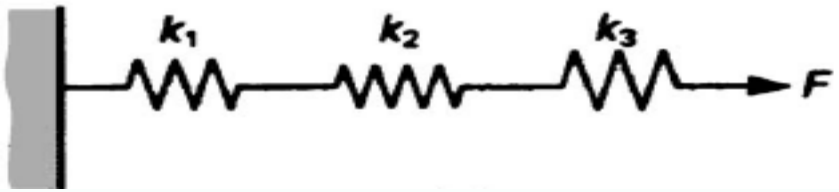
For a damper

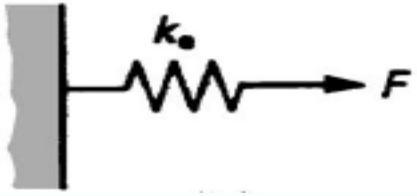
- If the fluid flow rate is small, the flow is **laminar** and the force is proportional to the velocity.
- Mechanical resistance that produces force proportional to the velocity is called **viscous friction**.
- If the fluid flow rate is large, the flow is **turbulent** and the force is proportional to the square of the velocity.
- The seal produces a constant friction force that is independent of velocity (Coulomb friction)
- Resistance due to both Coulomb friction and viscous friction is given by

$$F = F_c + cv$$



# Springs connected in series

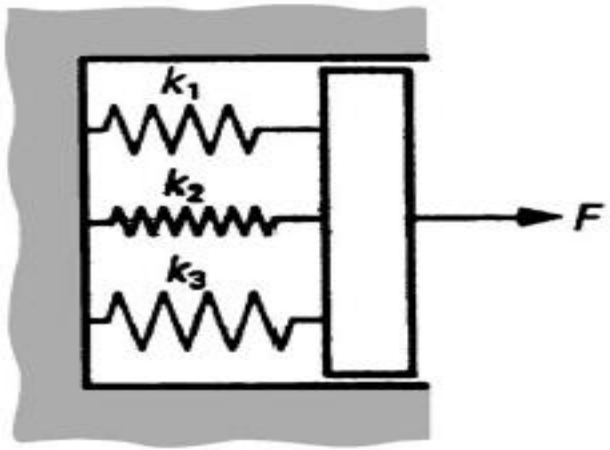


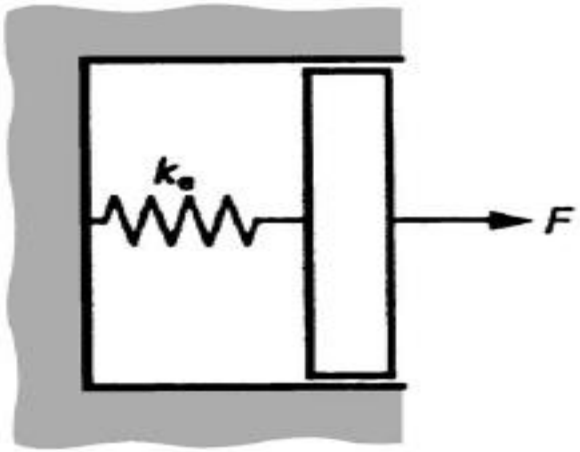




$$\delta = \delta_1 + \delta_2 + \delta_3$$

$$\frac{F}{k} = \frac{F}{k_1} + \frac{F}{k_2} + \frac{F}{k_3}; \qquad \frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

## Springs connected in parallel



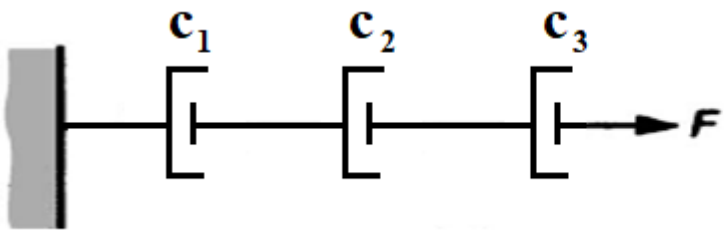



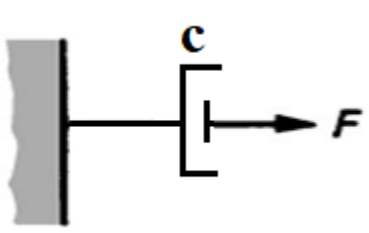

$$F = F_1 + F_2 + F_3$$

$$k\delta = k_1\delta + k_2\delta + k_3\delta$$

$$k = k_1 + k_2 + k_3$$

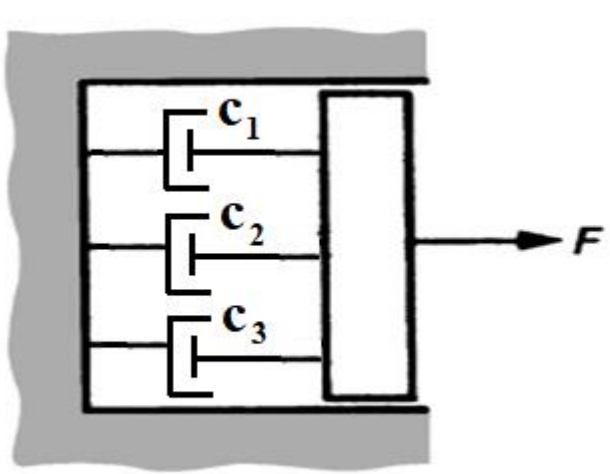
# Dampers connected in series

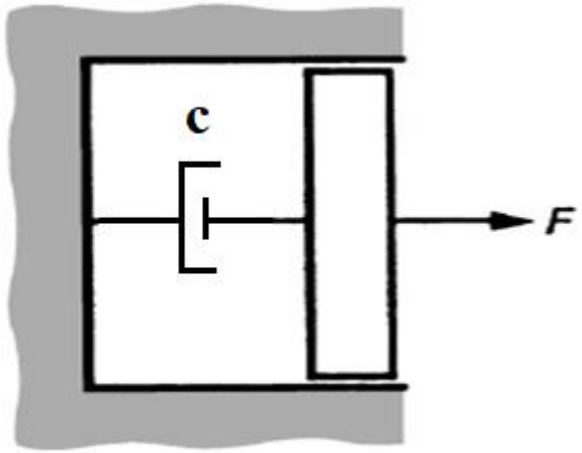



$$\frac{F}{c} = \frac{F}{c_1} + \frac{F}{c_2} + \frac{F}{c_3}; \qquad \frac{1}{c} = \frac{1}{c_1} + \frac{1}{c_2} + \frac{1}{c_3}$$

## Springs connected in parallel

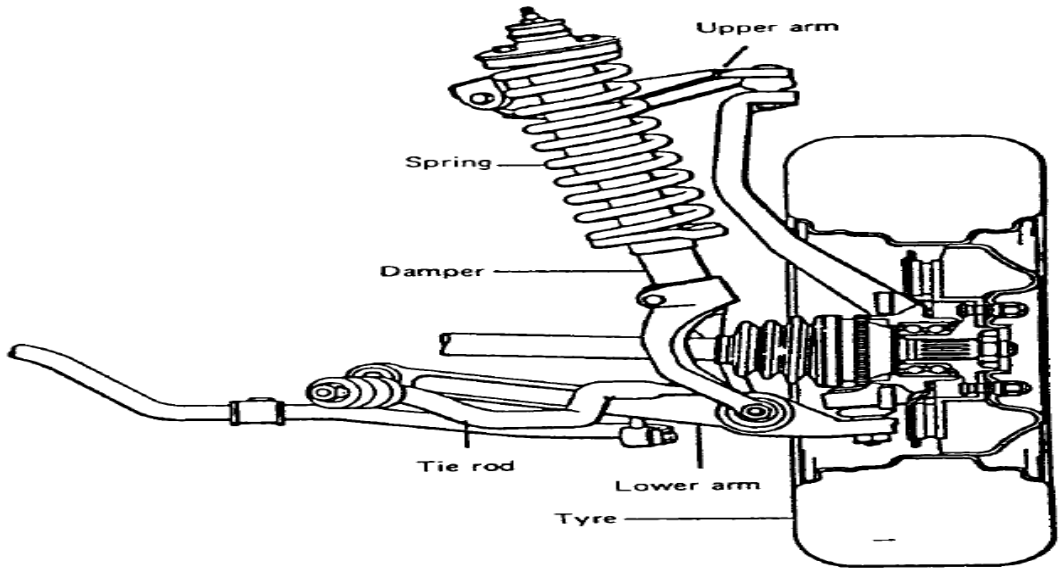


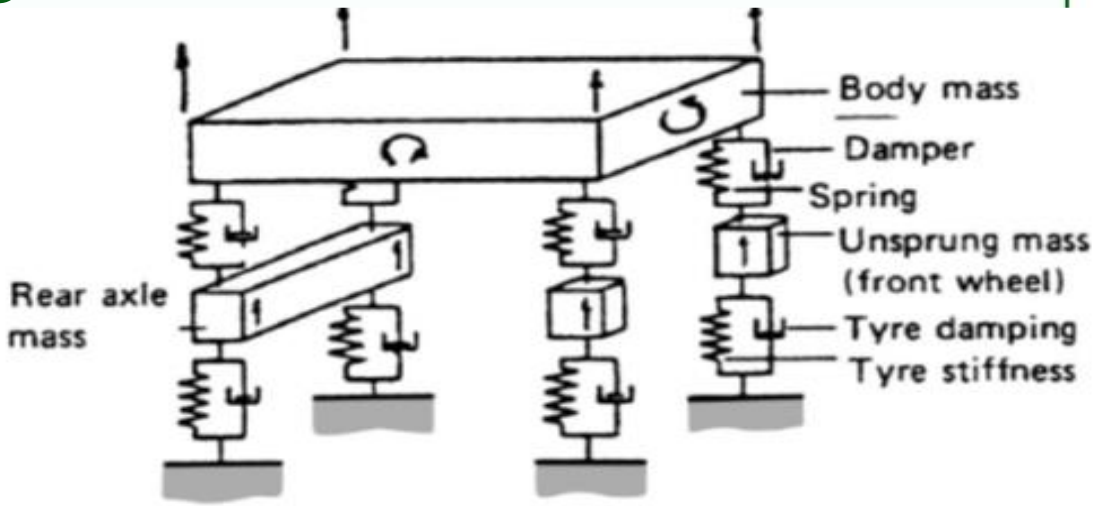

$$F = F_1 + F_2 + F_3$$

$$c = c_1 + c_2 + c_3$$

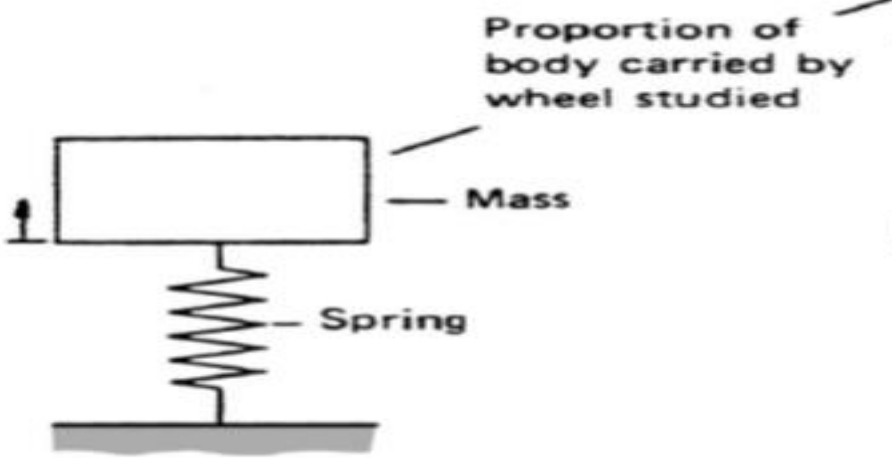
# Modeling



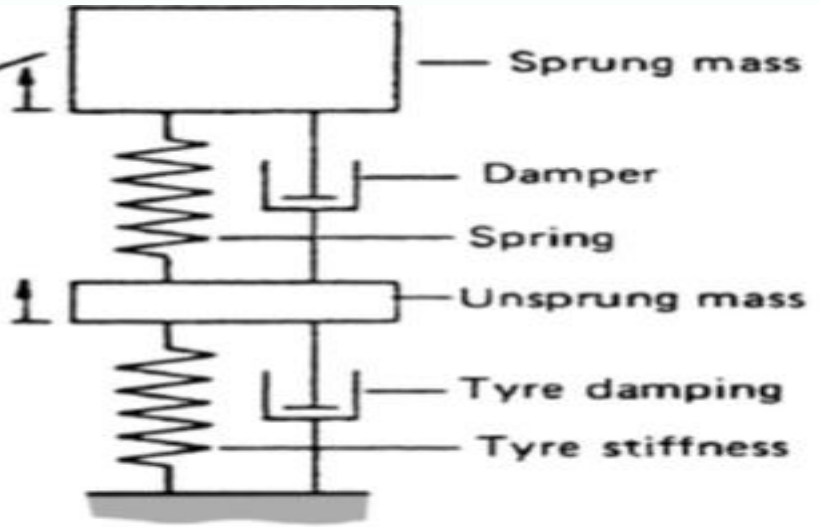

Rover 800 front suspension. (By courtesy of Rover Group.)




Motion in a vertical direction, roll, and pitch can be analysed

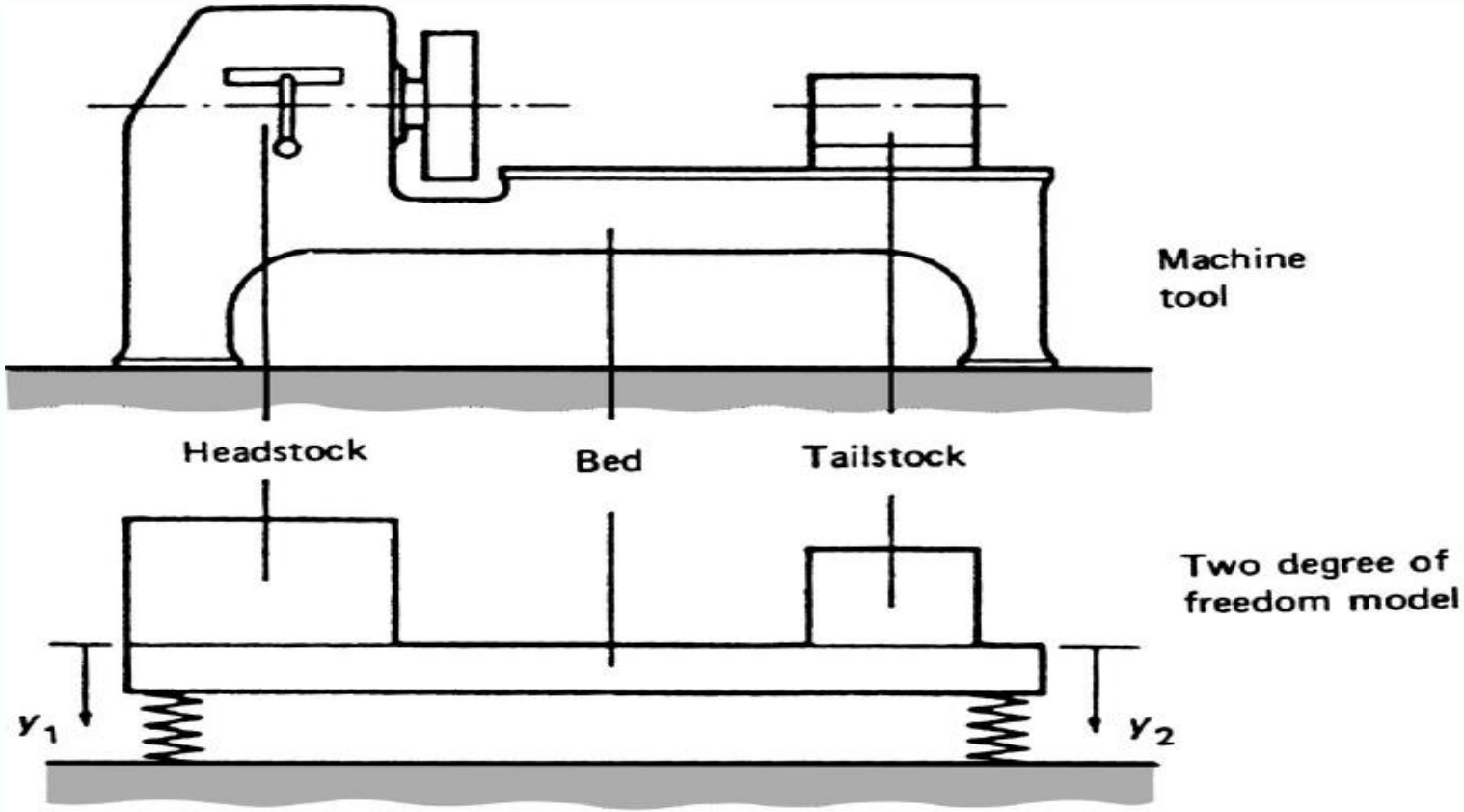



Simplest model – motion in a vertical direction only can be analysed.



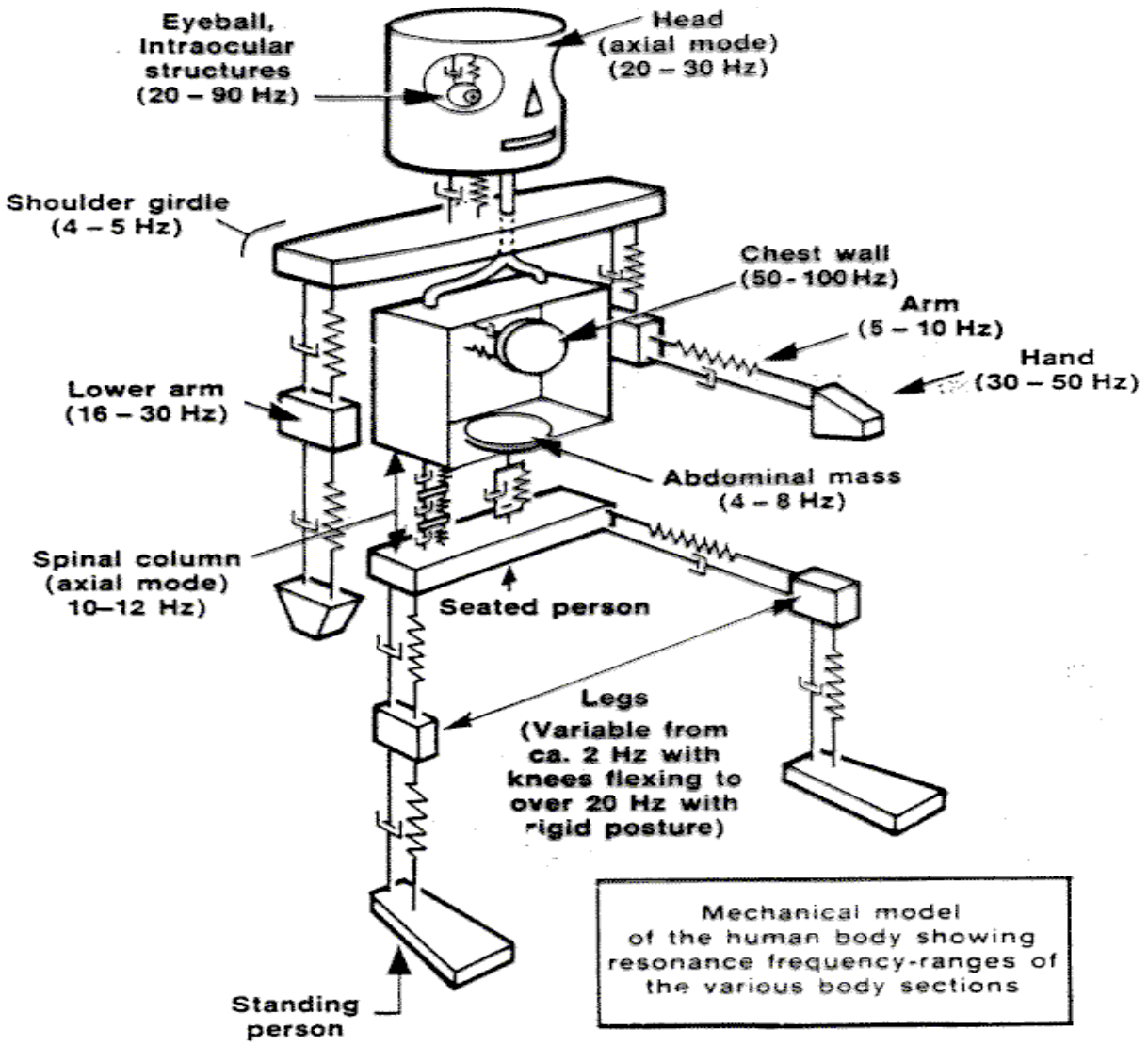

Motion in a vertical direction only can be analysed.

# Modeling




Machine tool vibration analysis model.

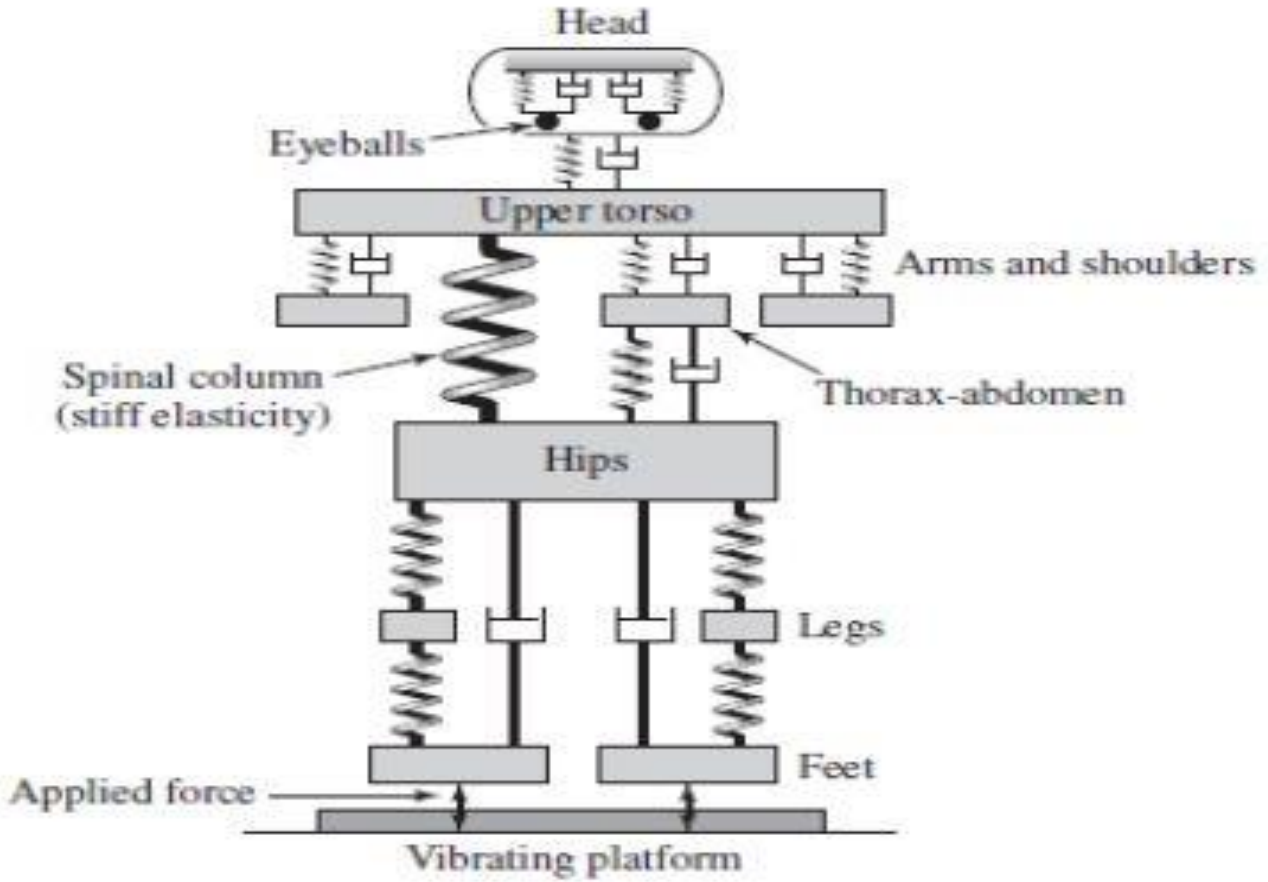
# Human body resonance frequencies



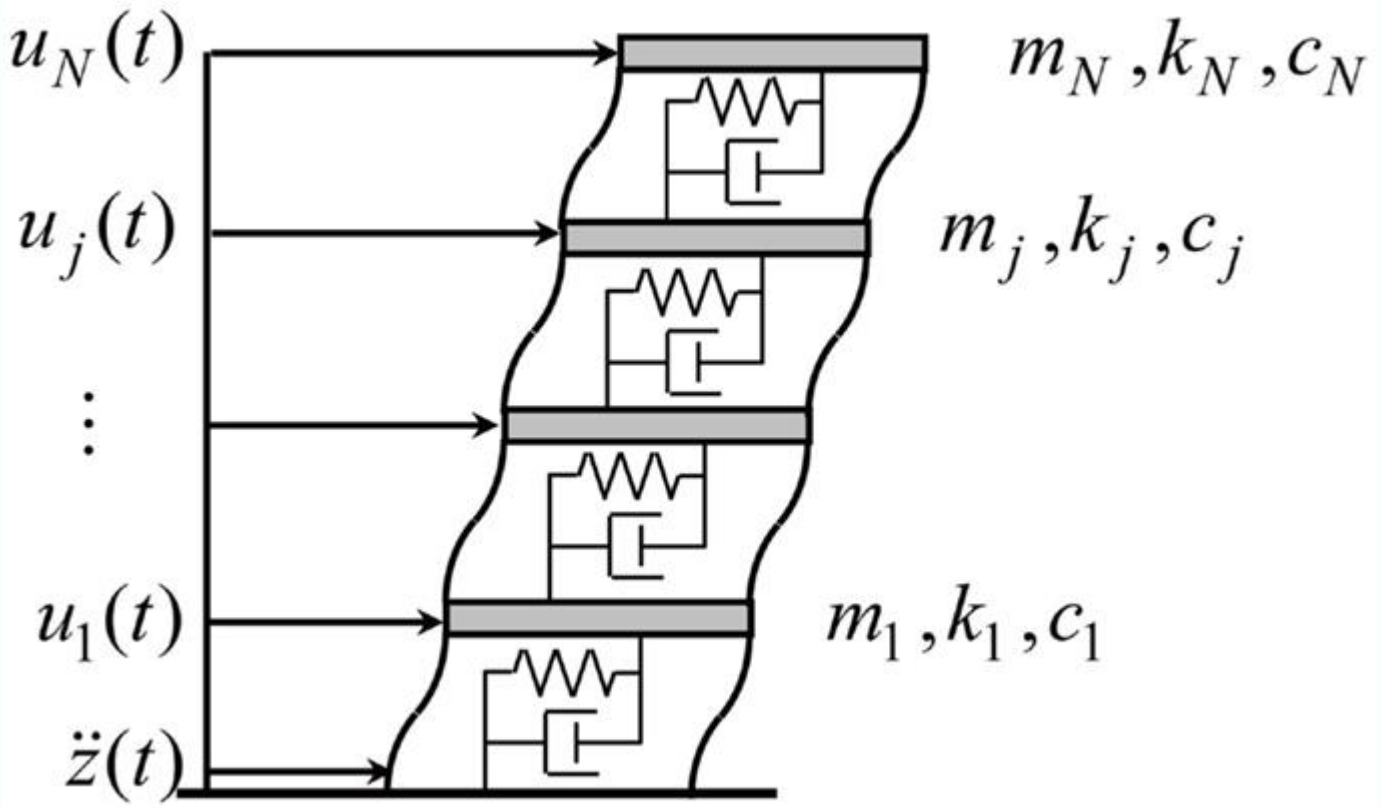

Mechanical model of the human body showing resonance frequency-ranges of the various body sections

# Human Model

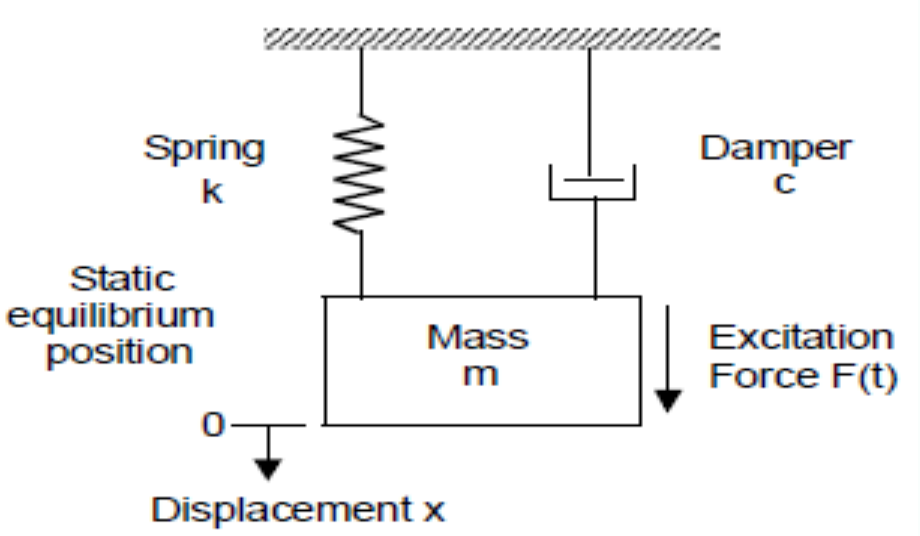

z
y
x


Head
Eyeballs
Upper torso
Arms and shoulders
Spinal column
(stiff elasticity)
Thorax-abdomen
Hips
Legs
Feet
Applied force
Vibrating platform

# Multistory Building Model



# EQUIVALENT SYSTEMS ANALYSIS




The kinetic energy of a linear system can be written in the form

$$K.E = \frac{1}{2}mv^2$$

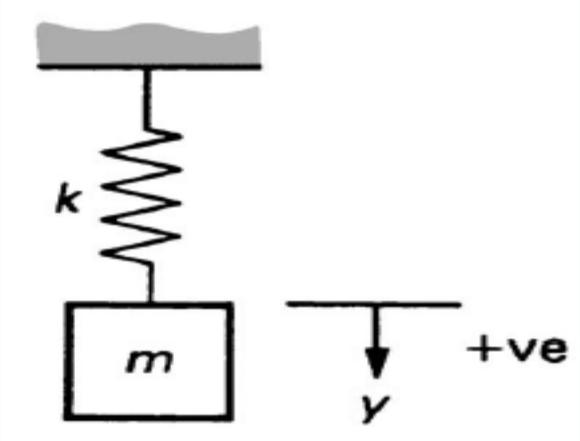
The potential energy of a linear system can be written in the form

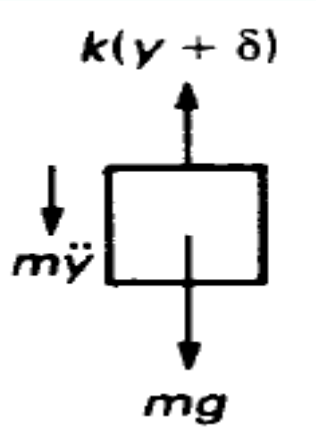
$$P.E = \int kx\, dx = \frac{1}{2}kx^2$$

The work done by the viscous damping force in a linear system between two arbitrary locations $x_1$, and $x_2$, can be written as

$$W = \int_{x_1}^{x_2} -cv\, dx$$

# FREE UNDAMPED VIBRATION-S.H.M




$$mg = k\delta$$

$$\frac{g}{\delta} = \frac{k}{m} = {\omega_n}^2$$

$$m\ddot{y} = -k(y + \delta) + mg;$$

$$\ddot{y} + \left(\frac{k}{m}\right)y = 0;$$

# Energy Method

- For undamped free vibration the total energy in the vibrating system is constant throughout the cycle. Therefore the maximum potential energy *V*, is equal to the maximum kinetic energy *T*, although these maxima occur at different times during the cycle of vibration.

$T + V =$ constant, and thus $\frac{d}{dt}(T + V) = 0.$

Example

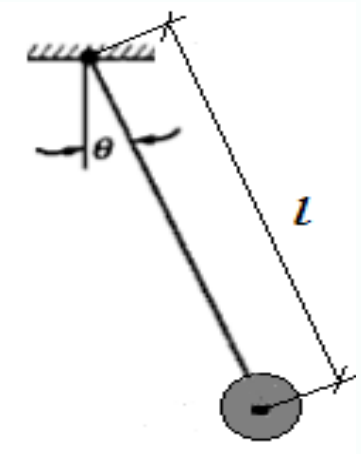
$$T = \frac{1}{2}m\dot{y}^2;\ V = \frac{1}{2}ky^2$$

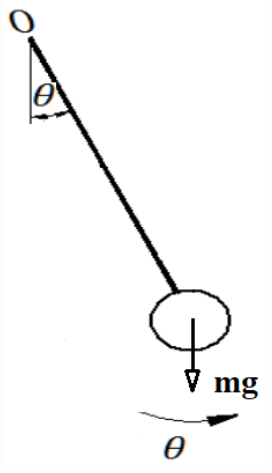
$$\frac{d}{dt}\left(\frac{1}{2}m\dot{y}^2 + \frac{1}{2}ky^2\right) = 0$$

$$m\ddot{y} + ky = 0$$

# The Simple Pendulum

## Newtonian Approach




$$\sum M = I\alpha = I\ddot{\theta}$$

$$-mglsin\theta = I\ddot{\theta};$$

$$I\ddot{\theta} + mgl\theta = 0$$

$$\ddot{\theta} + \omega^2\theta = 0;\ \ since\ I = ml^2$$

## Energy Method

$$P.E = V = mgl(1 - \cos\theta)$$

$$K.E = T = \frac{1}{2}I\dot{\theta}^2$$

$$\frac{d}{dt}(V + T) = 0$$

$$\frac{d}{dt}\left(mgl(1 - \cos\theta) + \frac{1}{2}I\dot{\theta}^2\right) = 0$$

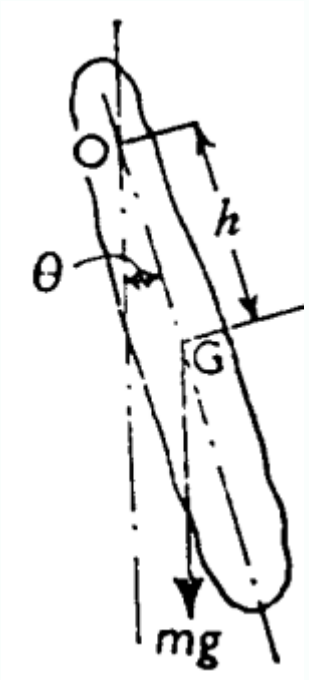
$$I\ddot{\theta} + mgl\theta = 0$$

$$\ddot{\theta} + \omega^2\theta = 0;\ \ since\ I = ml^2$$

$$where\ \ \omega^2 = \frac{g}{l}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}$$

# The Compound Pendulum




$$\sum M = I\alpha = I\ddot{\theta}$$

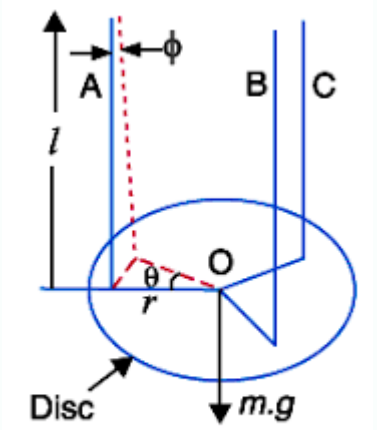
$$-mghsin\theta = I_o\ddot{\theta};$$

$$I_o\ddot{\theta} + mgh\theta = 0$$

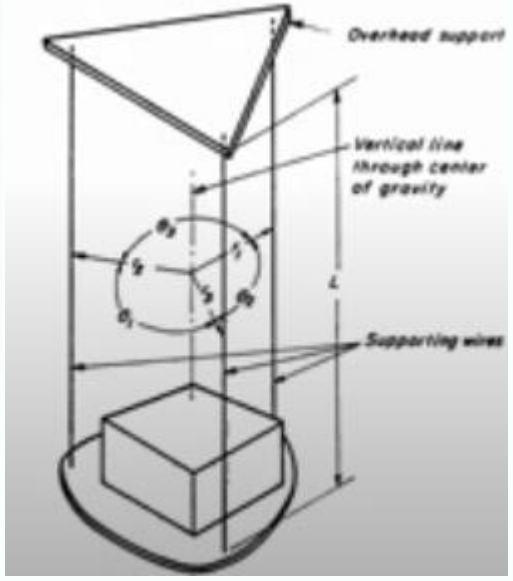
$$\ddot{\theta} + \omega^2\theta = 0;\ \ since I_o = I_G + mh^2$$

# Bifilar/Trifillar Suspension






Trifilar suspension.




Weight acting on each support mg/n

The horizontal component of weight acting on each support is

**(mg/n)sinϕ**

Moment of weight acting on each support is

**(mg/n)rsinϕ**

Total moment of weight acting on each support is

**n(mg/n)rsinϕ =mgrsinϕ**

$$\sum M_o = I_o = I_o\ddot{\theta}$$

$$-mgrsin\ \phi = I_o\ddot{\theta}$$

For small values of ϕ

$$I_o\ddot{\theta} + mgr\phi = 0$$

Since $r\theta = l\phi$

$$I_o\ddot{\theta} + \frac{mgr^2}{l} = 0$$

$$2\pi f = \sqrt{\frac{mgr^2}{I_o l}}$$

# Spring-mass system in which the mass of the spring matters

**Assume the mass $M$ has an amplitude of $X$**

**The kinetic energy of the mass *M shown in* the Figure is as before**

$$\frac{1}{2}M(\omega_n X)^2$$

**The mass of the element *δl* is $m\frac{\delta l}{L}$, *so* its kinetic energy at the mid position of the element is $\frac{1}{2}\left(m\frac{\delta l}{L}\right)(\omega_n \frac{l}{L}X)^2$, and of the whole spring is therefore**

$$\frac{1}{2}m{\omega_n}^2X^2\frac{1}{L^3}\int_0^L l^2dl = \frac{1}{2}\frac{m}{3}{\omega_n}^2X^2$$

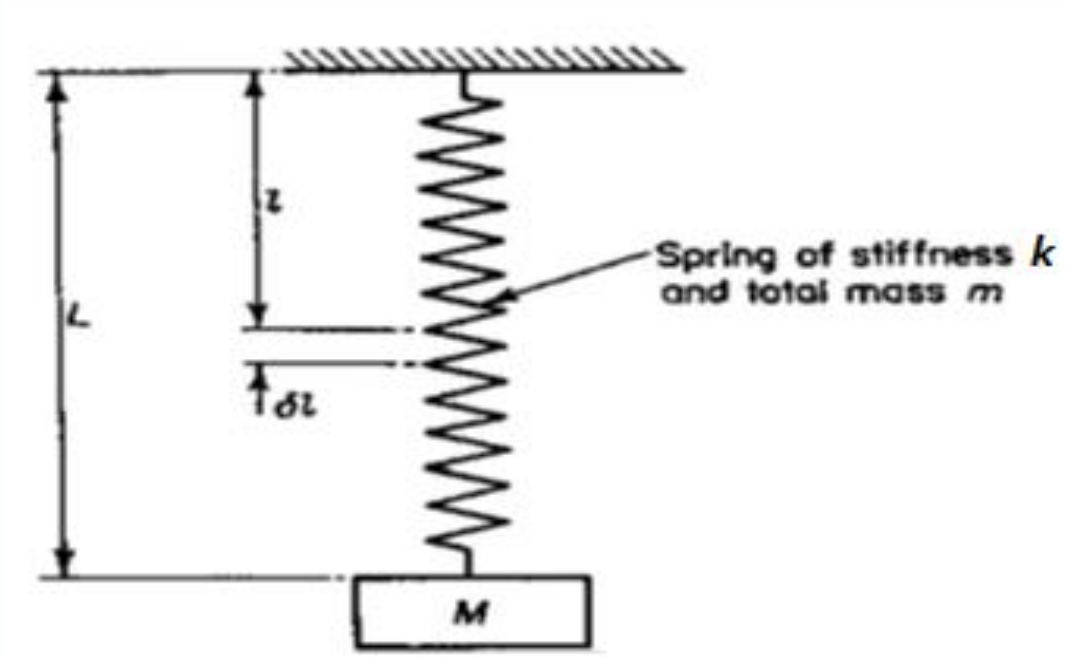
**So the total kinetic energy -in the system is**

$$\frac{1}{2}\left(M+\frac{m}{3}\right){\omega_n}^2X^2$$

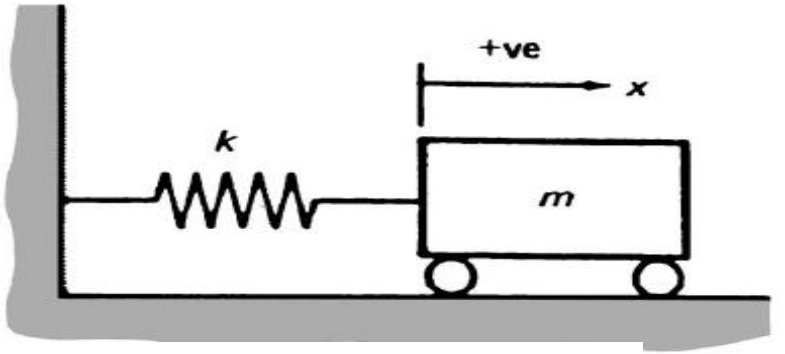
**and equating this to the strain energy we get**

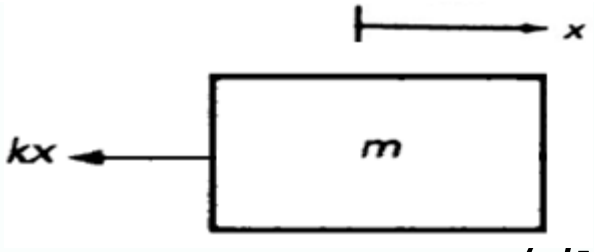
$$\omega_n = \sqrt{\frac{k}{M+\frac{1}{3}m}}\ rad/s$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{M+\frac{1}{3}m}}\ Hz$$



# FREE UNDAMPED VIBRATION




$$m\ddot{x} + kx = 0; \quad \ddot{x} + \left(\frac{K}{m}\right)x = 0;$$

$$\ddot{x} + {\omega_n}^2 x = 0; {\omega_n}^2 = \frac{k}{m}; f = \frac{\omega_n}{2\pi};$$

$$T = 1/f$$

$$x = A sin\omega_n t + B cos\omega_n t$$

or

$$x = C sin(\omega_n t + \emptyset)$$

Assumed initial conditions

$$x(0) = x_o; \ \dot{x}(0) = v_o$$

$$@\ t = 0; x = x_o$$

$$x_o = B$$

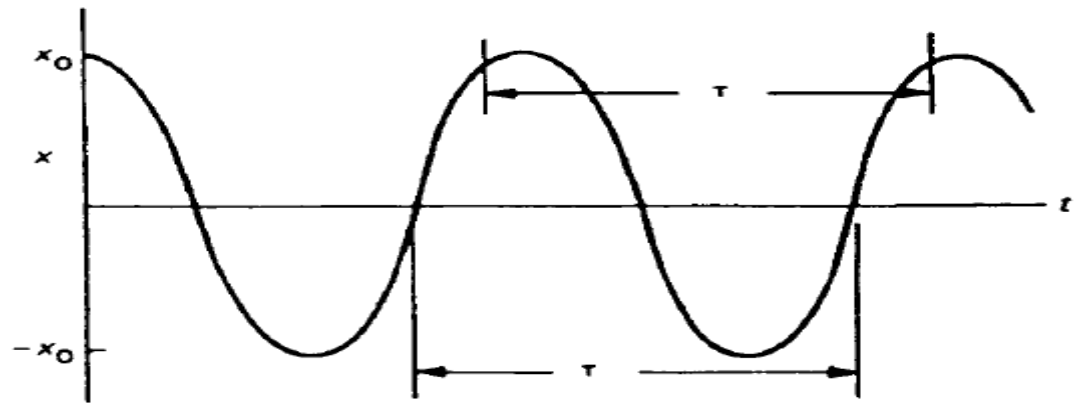
$$@\ t = 0; \dot{x} = v_o$$

$$\dot{x} = A\omega cos\omega_n t - B\omega sin\omega_n t$$

$$v_o = A\omega_n$$

$$A = v_o/\omega_n$$

$$x = (v_o/\omega_n) sin\omega_n t + x_o cos\omega_n t$$



# Kinematic quantities of SHM

$$x(t) = Csin(\omega_n t + \emptyset)$$

$$v(t) = C\omega_n cos(\omega_n t + \emptyset) = \omega_n \sqrt{C^2 - x^2}$$

$$a(t) = -{\omega_n}^2 x$$

# Example 1-1

Consider a small spring about 30 mm long, welded to a stationary table (ground) so that it is fixed at the point of contact, with a 12-mm bolt welded to the other end which is free to move. The mass of this system is about 49.2 g. The spring stiffness is measured to be $k$ = 857.8 N/m. Calculate the natural frequency and period. Also determine the maximum amplitude of the response if the spring is initially deflected 10 mm.

**Solution**

The natural frequency is

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{857.8\ N/m}{49.2\ x\ 10^{-3}\ kg}} = 132\ \text{rad/s}$$

In hertz, this becomes

$$f_n = \frac{\omega_n}{2\pi} = 21\ \text{Hz}$$

The period is $T = \frac{2\pi}{\omega_n} = \frac{1}{f_n} = 0.0476\ \text{s}$

# Solution

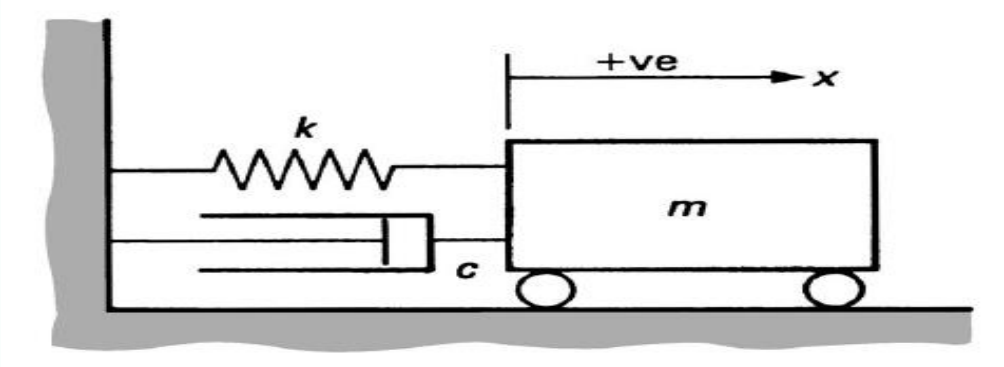
The maximum value of the displacement response,, corresponds to the value of the constant *A*. Assuming that no initial velocity is given to the spring ($v_o = 0$),

$$x(t)_{max} = A = \frac{\sqrt{\omega_n^2 x_o^2 + v_o^2}}{\omega_n} = x_o = 10\ \text{mm}$$

The maximum value of the velocity response is $\omega A\ or\ \omega x_o$ = 1320 mm/s and the acceleration response has maximum value $\omega_n^2 A = \omega_n^2 x_o$ = 174.24 X $10^3$ mm/$s^2$. Since $v_o$= 0, the phase is $\phi$ = $\tan^{-1}(\omega_n x_o/0)$ = π/2, or 90°. Hence, in this case, the response is

$$x(t) = 10\sin\left(132t + \frac{\pi}{2}\right) = 10\cos\ (132t)\ \text{mm}$$

# FREE DAMPED VIBRATION




$$m\ddot{x} + c\dot{x} + kx = 0$$

$$x = X_o e^{\lambda t}$$

$$\dot{x} = \lambda X_o e^{\lambda t}; \quad \ddot{x} = \lambda^2 X_o e^{\lambda t}$$

$$m\lambda^2 + c\lambda + k = 0$$

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

$$d = c^2 - 4mk$$

$d = 0;$ *Critically damped system*

$d > 0;$ Overdamped system

$d < 0;$ Less damped – underdamped System

$$d = c^2 - 4mk = 0$$

Critical damping; $c_c = 2\sqrt{mk} = 2m\omega_n$

Damping ratio, $\xi = \frac{c}{c_c} = \frac{c}{2m\omega_n}$

$\Rightarrow \xi = 1;$ *Critically damped system*

$\xi > 1;$ Overdamped system

$\xi < 1;$ Less damped – underdamped System

$$\lambda = \frac{-2\xi m\omega_n \pm \sqrt{(2\xi m\omega_n)^2 - (2m\omega_n)^2}}{2m}$$

$$\lambda = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

# FREE DAMPED VIBRATION

- $\xi = 1$; *Critically damped system*

$$\lambda_1 = \lambda_2 = -\omega_n$$

$$x(t) = (A + Bt)e^{-\omega_n t}$$

- $\xi > 1$; Overdamped system

$$\lambda_1 = -\xi\omega_n - \omega_n\sqrt{\xi^2 - 1}$$

$$\lambda_2 = -\xi\omega_n + \omega_n\sqrt{\xi^2 - 1}$$

$$x(t) = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$$

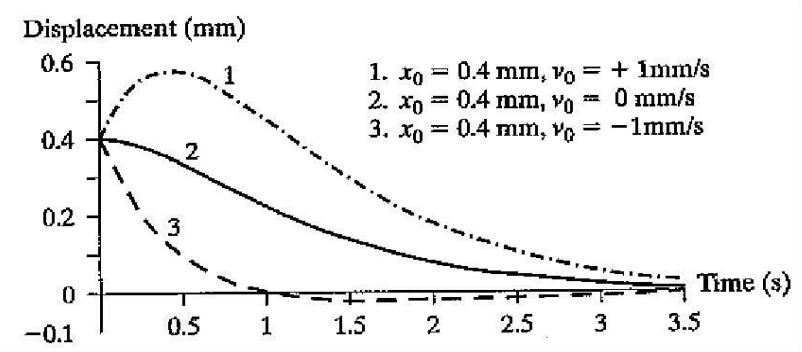
- $\xi < 1$; Less damped – underdamped System

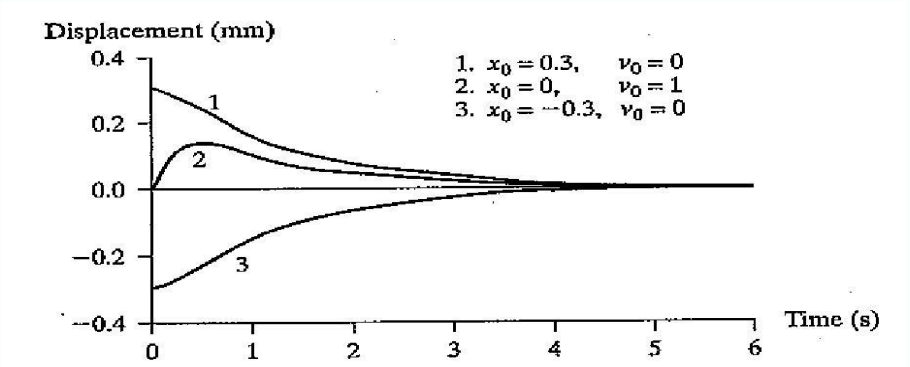
$$\lambda_1 = -\xi\omega_n - \omega_n j\sqrt{1 - \xi^2}$$

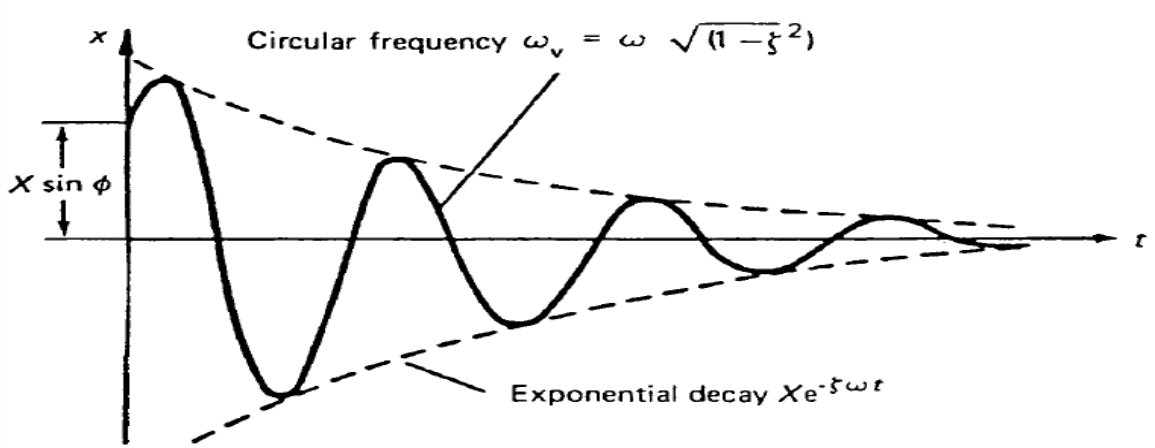
$$\lambda_2 = -\xi\omega_n + \omega_n j\sqrt{1 - \xi^2}$$

$$x(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

$\omega_d = \omega_n\sqrt{1 - \xi^2}$; Damped Natural Frequency



# Example 1-2

The damping rate of the spring is measured to be 0.11 kg/s. Calculate the damping ratio and determine if the free motion of the spring-bolt system( Example 1-1) is over-damped, under-damped, or critically damped. $m$ = 49.2 g and $k$ = 857.8 N/m

**Solution.**

Using the definition of the critical damping coefficient of equation

$$c_{cr} = 2\sqrt{km} = 2\sqrt{(2857.8\ \mathrm{N/m})(49.2 \times 10^{-3}\mathrm{kg}}$$

= 12.993 kg/s

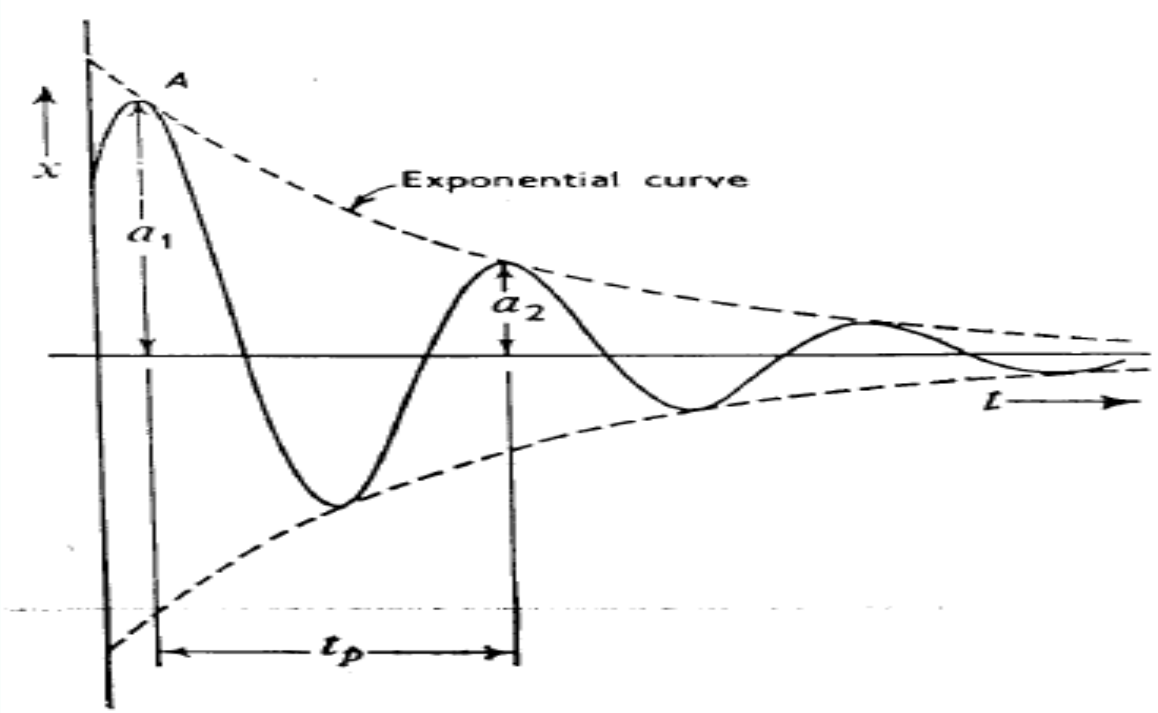
If $c$ is measured to be 0.11 kg/s, the critical damping ratio becomes

$$\xi = \frac{c}{c_{cr}} = \frac{0.11\ (\mathrm{kg/s})}{12.993\ (\mathrm{kg/s})} = 0.0085$$

or 0.85% damping. Since $\xi$ is less than 1, the system is under-damped. The motion resulting from giving the spring-bolt system a small displacement will be oscillatory.

# Logarithmic decrement

- A convenient way of determining the damping in a system is to measure the rate of decay of oscillation.




$$a_1 = Ae^{-\xi\omega_n t}\sin(\omega_d t + \phi)$$

$$a_2 = Ae^{-2\xi\omega_n t}\sin(2\omega_d t + \phi)$$

$$a_r = Ae^{-r\xi\omega_n t}\sin(r\omega_d t + \phi)$$

$$t = t_p = \frac{2\pi}{\omega_d}; r = 1, 2, 3 \ldots$$

$$\sin(r\omega_d t + \phi) = \sin(\omega_d t + \phi)$$

$$\sin(2\pi + \phi) = \sin(2\pi r + \phi)$$

$$\frac{a_1}{a_r} = e^{\xi\omega_n t(r-1)}$$

$$ln\left(\frac{a_1}{a_r}\right) = \xi\omega_n t(r-1) = \frac{2\pi\xi(r-1)}{\sqrt{1-\xi^2}}$$

# Example 1-3

A static deflection test is performed on a less damped system with a mass of 2 kg and the stiffness is determined to be 1.5 X $10^3$ N/m. The displacements at $t_1$ and $t_2$ are measured to be 9 and 1 mm, respectively. Calculate the damping coefficient.

## Solution

From the definition of the logarithmic decrement

$$\delta = \ln\left[\frac{x(t_1)}{x(t_2)}\right] = \ln\left[\frac{9\text{mm}}{1\text{mm}}\right] = 2.1972$$

From equation (1.65),

$$\xi = \sqrt{\frac{2.1972^2}{4\pi^2 + 2.1972^2}} = 0.33 \text{ or } 33\%$$

Also

$$c_{cr} = 2\sqrt{km} = 2\sqrt{(1.5 \times 10^3 \text{N/m})(2 \text{ kg})} = 1.095 \times 10^2 \text{ kg/s}$$

and from equation (1.23) the damping coefficient becomes

$$c = c_{cr}\xi = (1.095 \times 10^2)(0.33) = 36.15 \text{ kg/s}$$

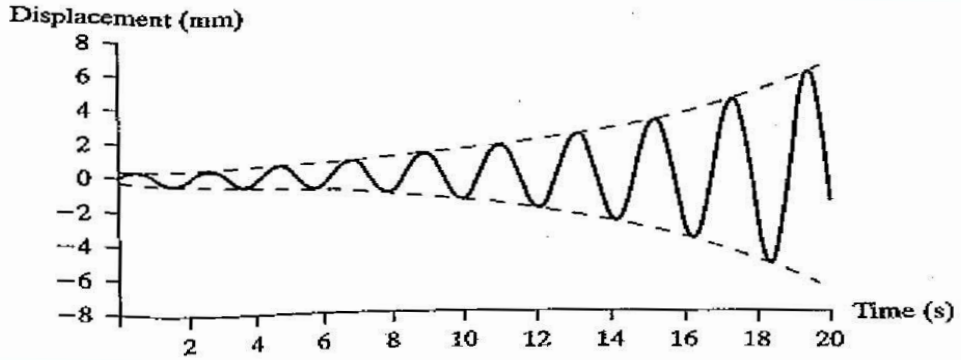
# Stability Considerations

- In cases where the vibration equation takes the form

$$m\ddot{x} - kx = 0$$

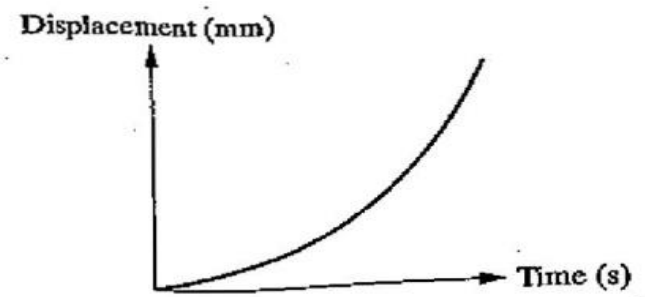
The solution takes the form

$$x(t) = A \sinh \omega_n t + B \cosh \omega_n t$$



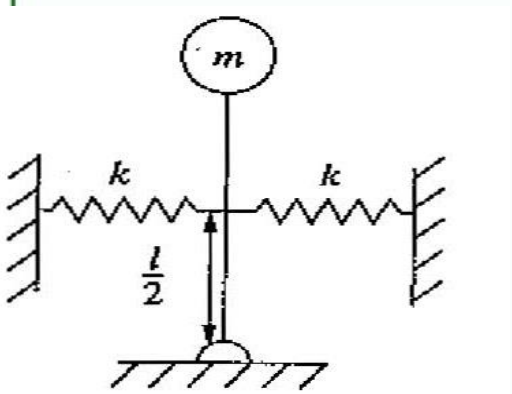

- $\sinh(x) = 0.5( e^{x} - e^{-x} )$ and $\cosh(x) = 0.5( e^{x} + e^{-x} )$

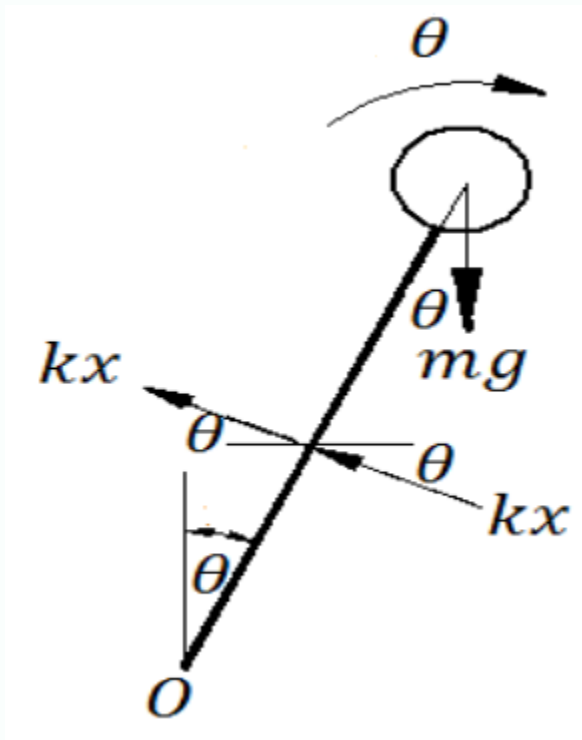
which increases without bound as *t* does. In this case $|x(t)|$ no longer remains finite and such solutions are called *divergent* or *unstable*.



# Example

Consider the inverted pendulum connected to two equal springs, shown. Assume that the springs are undeflected when in the vertical position and that the mass *m* of the ball at the end of the pendulum rod is substantially larger than the mass of the rod itself, so that the rod is considered to be massless. If the rod is of length $l$ and the springs are attached at the point $l/2$, the equation of motion becomes




$$ml^2\ddot{\theta} + \left(\frac{kl^2}{2}\sin\theta\right)\cos\theta - mgl\sin\theta = 0$$

For $\theta <$ about $\pi/20$, $\sin\theta \cong \theta$ and $\cos\theta \cong 1$. $ml^2\ddot{\theta} + \left(\frac{kl^2}{2}\theta\right) - mgl\theta = 0$

which upon rearranging becomes

$$2ml\ddot{\theta}(t) + (kl - 2mg)\theta(t) = 0$$

where $\theta$ is now restricted to be small (smaller than $\pi/20$). If *k*, *l*, and *m* are all such that the coefficient of $\theta$, called the effective stiffness, is negative, that is, if

$$kl - 2mg < 0; \text{ system unstable}$$